

Sports Analysis: Olympic Games

A Comprehensive Historical Analysis of Games, Athletes, and Medals

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Problem Statement & Scope

- **Objective:** Gain valuable insights into historical Olympic data by exploring games, sports, events, participants, medals, and regional representation.
- **Scope:** Analyze data from multiple editions to identify trends, patterns, and significant insights.
- **Primary Goal:** Provide a detailed overview of the Olympics' history and evolution.
- **Deliverable:** Comprehensive report and presentation showcasing findings and recommendations to enhance future Olympics' organization, sports selection, and regional representation.

Dataset Description: Table Explanations

- **Sport Table:** Comprehensive list of all sports (Summer/Winter), categorized into events.
- **Event Table:** Detailed event info, categorized by gender (Men's, Women's, Mixed).
- **City Table:** Cities that have hosted or participated in the Olympics.
- **Games Table:** Records every instance since 1896 (Year, Season).
- **Games City Table:** Joining table for Games and Cities (e.g., 1956 Melbourne/Stockholm).
- **NOC Region Table:** NOC codes mapped to participating countries.
- **Person Table:** Competitor info (Name, Gender, Height, Weight).
- **Person Region Table:** Links individuals to represented countries (accommodates country changes).
- **Games Competitor Table:** Links individuals to specific Olympic Games.
- **Medal Table:** Medal types (Gold, Silver, Bronze, N/A).
- **Competitor Event Table:** Largest table; links competitors, events, and medals.

Entity-Relationship (ER) Diagram

- The database structure is established through an Entity-Relationship Diagram (ERD).
- Cardinality and table relations are defined as 1-to-Many or Many-to-1.
- Cross filter directions are configured for better understanding of table structures and their relationships.

From: table (column) ↑	Relationship	To: table (column)
competitor_event (competitor...		games_competitor (id)
competitor_event (event_id)		event (id)
competitor_event (medal_id)		medal (id)
event (sport_id)		sport (id)
games_city (city_id)		city (id)
games_city (games_id)		games (id)
games_competitor (games_id)		games (id)
games_competitor (person_id)		person (id)
person_region (person_id)		person (id)
person_region (region_id)		noc_region (id)

MECE Breakdown of Analysis Areas

1. **Games Evolution:** Growth, editions, host cities, expansion.
2. **Sports & Events:** Structure, diversity, new/discontinued, gender events.
3. **Athlete Demographics:** Gender, age, height, weight, participation trends.
4. **Country Performance:** Country-wise participation, success, dominance.
5. **Medal Analysis:** Distribution patterns across countries, sports, genders.
6. **Geographic/Host Analysis:** Hosting patterns, regional representation.
7. **Participation vs Performance:** Athlete volume vs. medal efficiency.
8. **Gender & Diversity:** Inclusion trends, male/female/mixed representation.
9. **Historical Trends:** Long-term changes in participation and sports growth.
10. **Predictive Insights:** Future-oriented observations using historical data.

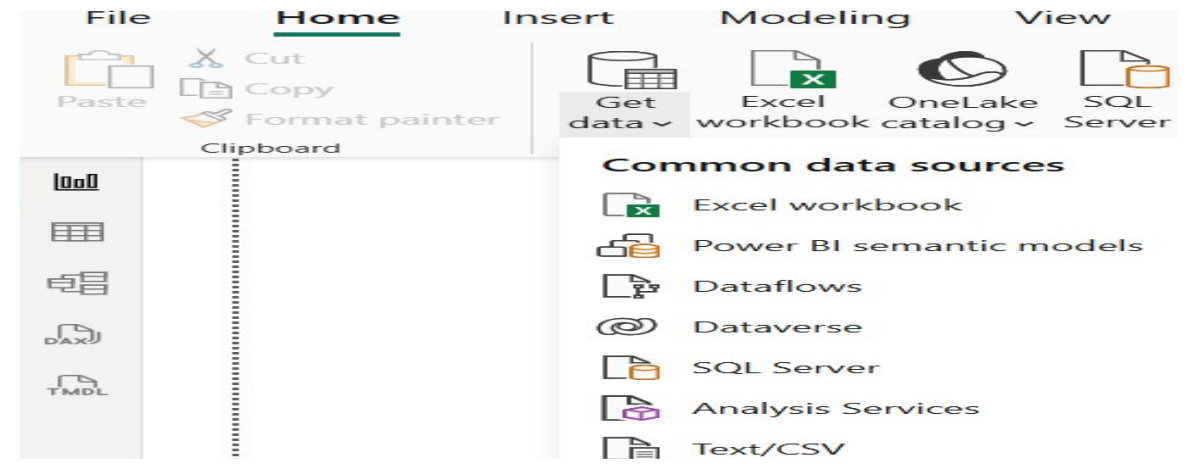
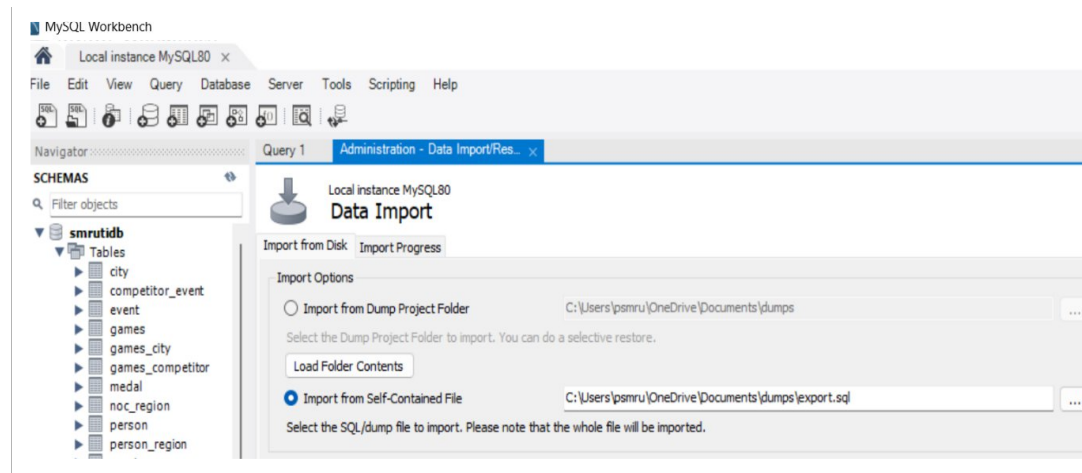
Data Connection and Access

- **MySQL Workbench (SQL Analysis):**

- Path: Server > Import Data > Import From Self Contained File.
- Imported SQL scripts of the whole dataset.

- **Power BI Dashboard:**

- Path: Get Data > Text/CSV.
- Imported CSV files for visual analysis.



EDA: Games & Hosting Trends (MySQL)

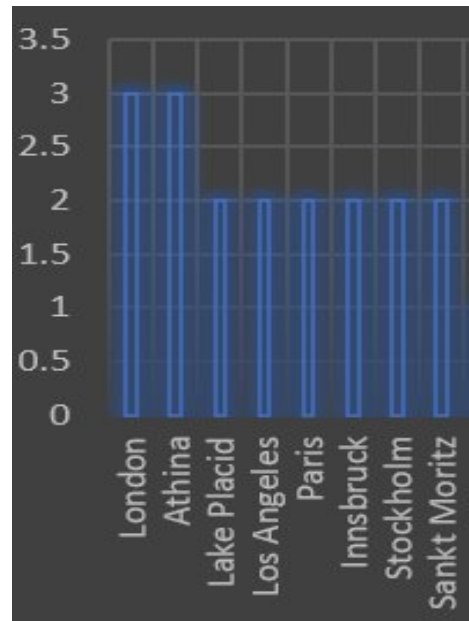
1. Frequency of Hosting:

- London and Athina hosted max times (3 each).
- Lake Placid, Los Angeles, Paris, Innsbruck, Stockholm, Sankt Moritz hosted 2 times.

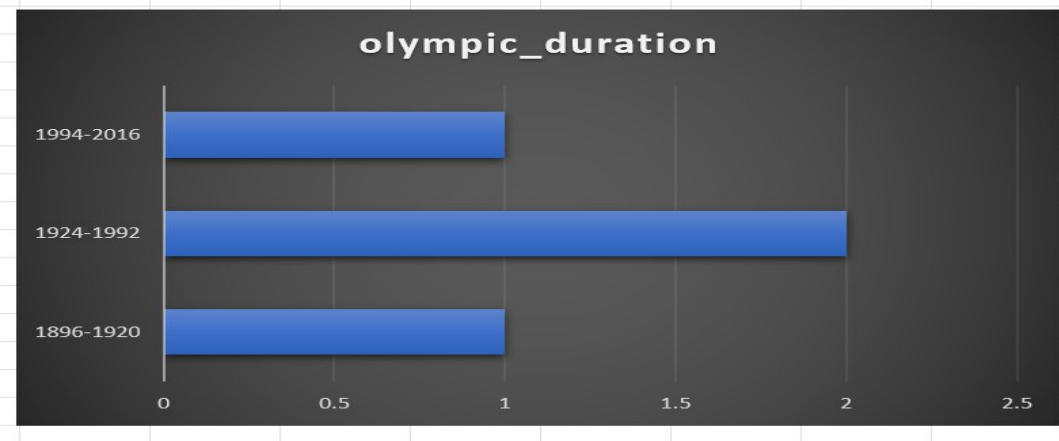
2. Duration of Games Over Time:

- 1896-1920 & 1994-2016: Single editions held.
- 1924-1992: Two Olympic events were held (Summer & Winter in same year).

```
select c.city_name, count(*) number_of_time_hosted
  from games g
    join games_city gc
  on g.id=gc.games_id
    join city c
  on gc.city_id=c.id
 group by c.city_name
 order by number_of_time_hosted desc;
```



```
select games_year, count(*) olympic_duration
  from games
 group by games_year
 order by games_year;
```



EDA: Notable & Emerging Events (MySQL)

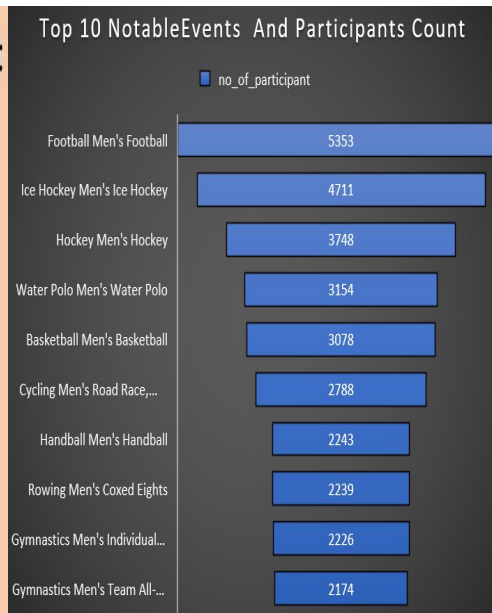
3. Notable Events (Highest Participants):

- Men's Football (5353), Men's Ice Hockey (4711), Men's Hockey (3748).
- Low participation: Gymnastic Men's Individual/Team (~2200).

4. Emerging Sports:

- Rugby Sevens introduced recently with ~300 participants across multiple events.

```
select e.event_name,count(ce.competitor_id) no_of_participant
from competitor_event ce
join event e
on ce.event_id=e.id
join medal m
on ce.medal_id=m.id
group by e.event_name
order by no_of_participant desc;
```



```
select s.sport_name,min(g.games_year) year_of_introduce,
count(ce.competitor_id) participants_per_sport
from competitor_event ce
join games_competitor gc
on ce.competitor_id=gc.id
join games g
on gc.games_id=g.id
join event e
on e.id=ce.event_id
join sport s
on e.sport_id=s.id
group by s.sport_name
order by year_of_introduce desc;
```

sport_name	year_of_introduce	participants_per_sport
Rugby Sevens	2016	299

EDA: Popularity Changes & Regional Sports (MySQL)

5. Sports that Lost Popularity (Significant drop 1896-1956 vs 1957-2016):

- Aeronautics, Alpinism, Art Competitions, Basque Pelota, Cricket, Croquet, Jeu De Paume, Lacrosse, Military Ski Patrol, Motorboating, Polo, Racquets, Roque, Rugby, Tug-Of-War.

6. Region/Culture-Specific Sports:

- Athletics is uniquely associated with Newfoundland, North Borneo, Somalia, South Sudan, and Turks & Caicos Islands (distinct sport count = 1).

```
SELECT
    s.sport_name,
    COUNT(ce.competitor_id) AS total_competitors
FROM competitor_event ce
JOIN games_competitor gc
ON ce.competitor_id = gc.id
JOIN games g
ON gc.games_id = g.id
JOIN event e
ON e.id = ce.event_id
JOIN sport s
ON e.sport_id = s.id
WHERE g.games_year BETWEEN 1896 AND 1956
GROUP BY s.sport_name
ORDER BY sport_name;
```

```
SELECT
    s.sport_name,
    COUNT(ce.competitor_id) AS total_competitors
FROM competitor_event ce
JOIN games_competitor gc
ON ce.competitor_id = gc.id
JOIN games g
ON gc.games_id = g.id
JOIN event e
ON e.id = ce.event_id
JOIN sport s
ON e.sport_id = s.id
WHERE g.games_year BETWEEN 1957 AND 2016
GROUP BY s.sport_name
ORDER BY sport_name;
```

```
SELECT
    nr.region_name,
    MIN(s.sport_name) AS sport_name,
    COUNT(DISTINCT s.sport_name) AS
        distinct_sport_count
FROM sport s
JOIN event e ON s.id = e.sport_id
JOIN competitor_event ce ON e.id = ce.event_id
JOIN games_competitor gc ON gc.id = ce.competitor_id
JOIN person_region pr ON pr.person_id = gc.person_id
JOIN noc_region nr ON nr.id = pr.region_id
GROUP BY nr.region_name
HAVING COUNT(DISTINCT s.sport_name) = 1;
```

region_name	sport_name	distinct_sport_count
Newfoundland	Athletics	1
North Borneo	Athletics	1
Somalia	Athletics	1
South Sudan	Athletics	1
Turks and Caicos Islands	Athletics	1

EDA: Gender Disparity in Sports (MySQL)

7. Sports with higher events for one gender:

- Significant male dominance: Football (5353 M vs 1009 F), Ice Hockey (4711 M vs 754 F), Wrestling (6346 M vs 304 F), Boxing (5648 M vs 72 F).
- Female-only dominance: Softball (0 M vs 478 F), Rhythmic Gymnastics (0 M vs 658 F), Synchronized Swimming (0 M vs 903 F).
- Male-only historical: Tug-Of-War, Nordic Combined, Baseball, Lacrosse, Polo, Cricket, Racquets.

```
SELECT
    s.sport_name,
    SUM(CASE WHEN p.gender = 'M' THEN 1 ELSE 0 END) AS
        male_event_count,
    SUM(CASE WHEN p.gender = 'F' THEN 1 ELSE 0 END) AS
        female_event_count
    FROM person p
    JOIN games_competitor gc ON p.id = gc.person_id
    JOIN competitor_event ce ON gc.id = ce.competitor_id
    JOIN event e ON e.id = ce.event_id
    JOIN sport s ON s.id = e.sport_id
    GROUP BY s.sport_name
    HAVING male_event_count <> female_event_count
```

sport_name	male_event_count	female_event_count
Basketball	3078	1256
Judo	2668	1093
Football	5353	1009
Tug-Of-War	125	0
Speed Skating	3499	2077
Cross Country Skiing	5704	3385
Athletics	25790	11571
Ice Hockey	4711	754
Swimming	12869	9727
Badminton	717	740
Sailing	5347	925
Biathlon	3024	1863
Gymnastics	16451	9012
Art Competitions	2770	301
Alpine Skiing	5325	3387
Handball	2243	1387
Weightlifting	3262	463
Wrestling	6346	304
Luge	1096	377

EDA: New & Discontinued Events (MySQL)

8. Newly Introduced Events (Recent Editions):

- Rugby Sevens (Women's & Men's), Wrestling Women's Light-Heavyweight/Featherweight Freestyle, Sailing Women's Skiffs.

9. Discontinued/Removed Events:

- Multiple events omitted from recent editions.
- Methodology: CTE used to extract event names not present in recent games.

```
WITH event_years AS (  
    SELECT  
        e.event_name,  
        MIN(g.games_year) AS first_year  
    FROM games_competitor gc  
    JOIN competitor_event ce ON gc.id = ce.competitor_id  
    JOIN games g ON g.id = gc.games_id  
    JOIN event e ON e.id = ce.event_id  
    GROUP BY e.event_name  
)  
  
SELECT event_name, first_year  
FROM event_years  
WHERE first_year = 2016;
```

event_name	first_year
Rugby Sevens Women's Rugby Sevens	2016
Wrestling Women's Light-Heavyweight, Freestyle	2016
Wrestling Women's Featherweight, Freestyle	2016
Rugby Sevens Men's Rugby Sevens	2016
Sailing Women's Skiff	2016

```
(WITH GAME_YEAR AS  
    (SELECT  
        e.event_name,  
        MIN(g.games_year) AS first_year  
    FROM games_competitor gc  
    JOIN competitor_event ce ON gc.id = ce.competitor_id  
    JOIN games g ON g.id = gc.games_id  
    JOIN event e ON e.id = ce.event_id  
    GROUP BY e.event_name)  
    SELECT event_name omitted_event, first_year  
    from GAME_YEAR  
    where first_year <> 2016  
    order by first_year desc;
```

Ommited_Events_From_Recent_Olympic

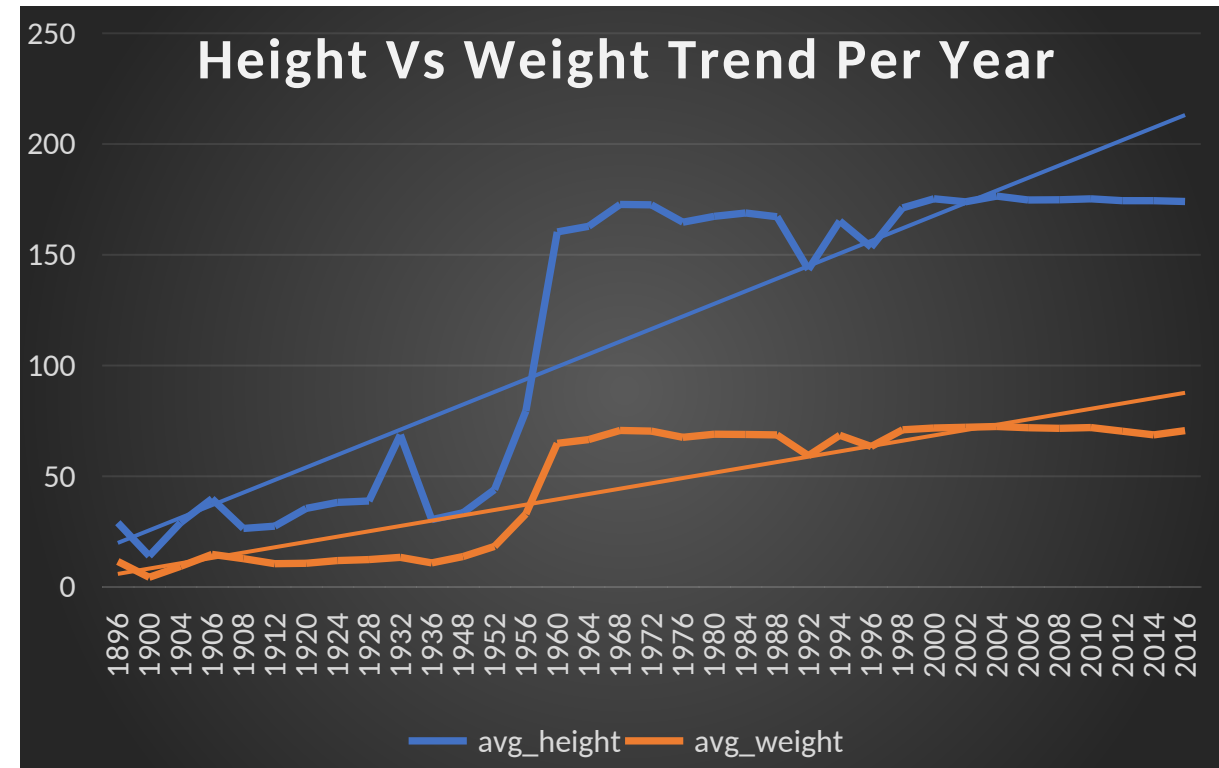
Figure Skating Mixed Team
Biathlon Mixed 2 x 6 kilometres and 2 x 7.5 kilometres Relay
Freestyle Skiing Men's Slopestyle
Freestyle Skiing Women's Halfpipe
Ski Jumping Women's Normal Hill, Individual
Snowboarding Women's Slopestyle
Snowboarding Men's Parallel Slalom
Luge Mixed Team Relay
Freestyle Skiing Women's Slopestyle
Snowboarding Men's Slopestyle

EDA: Athlete Demographics - Height & Weight (MySQL)

10. Trends in Height and Weight Over Time:

- Both average height and weight show a clear upward trend over time.
- Dotted trendlines confirm a positive linear progression.
- Athletes have generally become taller and heavier over the decades (e.g., avg height/weight jumped significantly from the 1960s onwards).

```
select g.games_year, avg(p.height) avg_height, avg(p.weight) avg_weight
      from games g
    join games_competitor gc
      on g.id=gc.games_id
    join person p
      on p.id=gc.person_id
  where p.height is not null and p.weight is not null
 group by g.games_year
 order by g.games_year;
```



EDA: Country Dominance in Specific Sports (MySQL)

11. Dominant Countries in Specific Sports:

- Sweden: Ice Hockey (386 medals)
- USA: Ice Hockey (376) & Art Competitions (296)
- India: Men's Hockey (307 medals) • Finland: Ice Hockey (323)

14. High medal counts from Specific Regions:

- Men's Ice Hockey dominated by Sweden, USA, Canada, Finland, Czechoslovakia, Switzerland.

```
select s.sport_name,
       e.event_name,
       nr.region_name,
       count(ce.medal_id) medal_count
from noc_region nr
join person_region pr
on nr.id=pr.region_id
join games_competitor gc
on gc.person_id=pr.person_id
join competitor_event ce
on ce.competitor_id=gc.id
join event e
on e.id=ce.event_id
join sport s
on e.sport_id=s.id
group by s.sport_name,e.event_name,nr.region_name
order by medal_count desc;
limit 10;
```

sport_name	event_name	region_name	medal_count
Ice Hockey	Ice Hockey Men's Ice Hockey	Sweden	386
Ice Hockey	Ice Hockey Men's Ice Hockey	USA	376
Ice Hockey	Ice Hockey Men's Ice Hockey	Canada	368
Ice Hockey	Ice Hockey Men's Ice Hockey	Finland	323
Hockey	Hockey Men's Hockey	India	307
Hockey	Hockey Men's Hockey	UK	303
Art Competitions	Art Competitions Mixed Painting, Unknown Event	USA	296
Ice Hockey	Ice Hockey Men's Ice Hockey	Czechoslovakia	291
Ice Hockey	Ice Hockey Men's Ice Hockey	Switzerland	279
Hockey	Hockey Men's Hockey	Netherlands	274

```
select nr.region_name,e.event_name,s.sport_name, count(ce.medal_id) medal_count
from noc_region nr
join person_region pr
on nr.id=pr.region_id
join games_competitor gc
on pr.person_id=gc.person_id
join competitor_event ce
on ce.competitor_id=gc.id
join event e
on e.id=ce.event_id
join medal m
on m.id=ce.medal_id
join sport s
on s.id=e.sport_id
group by nr.region_name,e.event_name,s.sport_name
order by medal_count desc
limit 10;
```

region_name	event_name	sport_name	medal_count
Sweden	Ice Hockey Men's Ice Hockey	Ice Hockey	386
USA	Ice Hockey Men's Ice Hockey	Ice Hockey	376
Canada	Ice Hockey Men's Ice Hockey	Ice Hockey	368
Finland	Ice Hockey Men's Ice Hockey	Ice Hockey	323
India	Hockey Men's Hockey	Hockey	307
UK	Hockey Men's Hockey	Hockey	303
USA	Art Competitions Mixed Painting, Unknown Event	Art Competitions	296
Czechoslovakia	Ice Hockey Men's Ice Hockey	Ice Hockey	291
Switzerland	Ice Hockey Men's Ice Hockey	Ice Hockey	279
Netherlands	Hockey Men's Hockey	Hockey	274

EDA: Success Factors & Consistent Performers (MySQL)

12. Factors Contributing to Success:

- Larger delegations = more medals overall; smaller nations = higher efficiency (medals/athlete).
- Sport diversity increases opportunities. Physical attributes show limited direct correlation vs. strategic focus.

13. Consistently Performing Countries:

- USA tops the list (max medals across multiple editions, e.g., 957 in 1992, 886 in 1988).
- Germany, UK, and Russia/Soviet Union also perform consistently well across decades.

```
select nr.region_name,  
count(distinct gc.id)total_participants,  
count(distinct e.sport_id) sports_count,  
count(ce.medal_id) medal_count,  
avg(p.height) avg_height,  
avg(p.weight) avg_weight  
from noc_region nr  
join person_region pr  
on nr.id=pr.region_id  
join games_competitor gc  
on gc.person_id=pr.person_id  
join competitor_event ce  
on ce.competitor_id=gc.id  
join event e  
on e.id=ce.event_id  
join person p  
on p.id=pr.person_id  
group by nr.region_name  
order by medal_count desc;
```

region	total_participants	sports_count	medal_count	avg_height	avg_weight
USA	12392	58	18127	141.4432	55.9176
UK	7999	59	11536	121.2031	48.5269
France	7323	57	11475	127.1771	48.9933
Germany	7119	53	10496	139.59	56.5142
Italy	6958	51	10355	136.5655	53.0769
Canada	6332	53	9540	149.2739	59.3966
Sweden	5416	47	8335	116.1969	46.9073
Japan	5355	52	8331	152.5606	56.9393
Australia	5452	51	7584	158.092	63.7766
Hungary	4045	43	6514	128.9467	52.5774
Poland	4241	45	6209	162.8156	65.9113

```
select nr.region_name,  
g.games_year,  
count(distinct gc.id)participant_count,  
count(ce.medal_id) medal_count  
from games g  
join games_competitor gc  
on g.id=gc.games_id  
join person_region pr  
on pr.person_id=gc.person_id  
join noc_region nr  
on nr.id=pr.region_id  
join competitor_event ce  
on ce.competitor_id=gc.id  
group by nr.region_name, g.games_year  
order by medal_count desc  
limit 50;
```

region_name	games_year	participant_count	medal_count
USA	1992	685	957
USA	1988	624	886
USA	1904	414	882
Germany	1992	581	866
Unified Team	1992	601	861
USA	1996	646	834
USA	1984	613	824
Soviet Union	1988	581	798
Australia	2000	620	793

EDA: Surprises, Growth & Geographic Factors (MySQL)

15. Surprising Medal Wins: Refugee Olympic Team participated 1st time recently and grabbed 12 medals.

16. Significant Growth/Decline: Australia, Unified Team, USA, China experienced notable changes in athlete participation count consecutively (Tracked via SQL LAG() window function).

17. Cultural/Geographical Factors: Countries with high participant counts AND good physical balance (BMI) grabbed max medals (USA, UK, France, Germany, Italy, Australia).

```
select nr.region_name,
min(g.games_year) game_yr,
count(ce.medal_id) medal_count
from games g
join games_competitor gc
on g.id=gc.games_id
join person_region pr
on pr.person_id=gc.person_id
join noc_region nr
on nr.id=pr.region_id
join competitor_event ce
on ce.competitor_id=gc.id
group by nr.region_name
order by game_yr desc;
```

```
with participation as
(select nr.region_name country,
g.games_year oly_year,
count(distinct gc.person_id) total_athlete
from games g
join games_competitor gc
on g.id=gc.games_id
join person_region pr
on pr.person_id=gc.person_id
join noc_region nr
on nr.id=pr.region_id
join competitor_event ce
on ce.competitor_id=gc.id
group by nr.region_name,g.games_year
order by nr.region_name,g.games_year),
trend as
(select country,
oly_year,
total_athlete current_year_athlete,
lag(total_athlete,1) over(partition by country order by oly_year) prev_year_athlete,
abs(total_athlete-lag(total_athlete,1) over(partition by country order by oly_year)) athlete_diff
from participation)

select *
from trend
order by athlete_diff desc
limit 5;
```

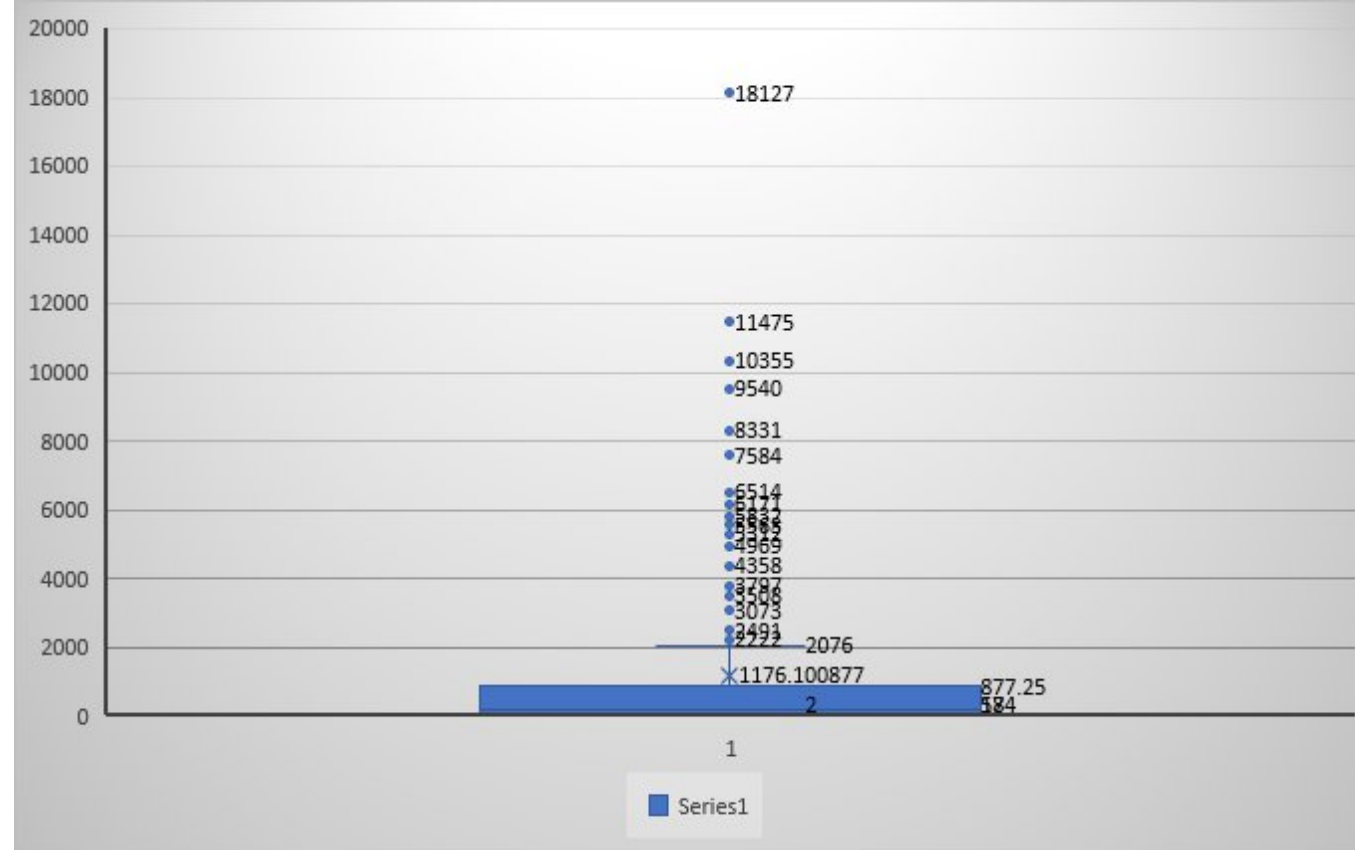
```
select nr.region_name,
s.sport_name,
count(distinct gc.person_id) total_participant,
avg(p.height) overall_height,
avg(p.weight) overall_weight,
10000*(avg(p.weight)/power(avg(p.height),2)) BMI,
count(ce.medal_id) medal_count
from competitor_event ce
join games_competitor gc
on ce.competitor_id=gc.id
join person_region pr
on pr.person_id=gc.person_id
join noc_region nr
on nr.id=pr.region_id
join event e
on e.id=ce.event_id
join sport s
on s.id=e.sport_id
join person p
on p.id=gc.person_id
group by nr.region_name,s.sport_name
order by medal_count desc
limit 10;
```


EDA: Impact on Overall Medal Tally (MySQL)

18. Regions with Notable Impact on Medal Tally:

- Created a box plot to identify outliers impacting the overall tally.
- Found 36 data points (countries) considered as outliers.
- Major impactful regions: USA, UK, France, Germany, Italy, Soviet Union, China, Australia, and 28 others.

Impactful_region
USA
UK
France
Germany
Italy
Canada
Sweden
Japan
Australia
Hungary
Poland
Soviet Union
Netherlands
Switzerland
Russia
Finland
Spain
China
Austria
Norway
Czechoslovakia
South Korea
Romania
West Germany
Brazil
Denmark
Bulgaria
Belgium
East Germany
Argentina
Ukraine
Mexico
Yugoslavia
Cuba
New Zealand
Greece



Power BI: Games & Season Overview

- 1. Summer vs Winter Games:** 29 Summer vs 22 Winter games (KPI Card + Slicer).
- 2. Games Across Decades:** Grouped games year by decade, displayed in a simple table visual.
- 4. Sports Distribution:** 52 Summer sports vs 17 Winter sports (KPI Card + Season Slicer).

season
■ Summer
□ Winter

29
Total_Games

season
□ Summer
■ Winter

22
Total_Games

season
■ Summer
□ Winter

52
Total_Sports

season
□ Summer
■ Winter

17
Total_Sports

Decades	Total_Games
1918-1928	2
1929-1939	2
1940-1950	1
1951-1961	3
1962-1972	3
1973-1983	2
1984-1994	4
1995-2005	2
2006-2016	3
Total	22

Power BI: Host Cities & Sports/Events

3. Top Host Cities: Table visual sorted descending by Total Games measure.

5. Highest Events by Sport: Athletics has the highest number of events (Table visual with Total Event measure).

7. Events Per Sport: Simple table visual mapping Sports Name to Total Event measure.

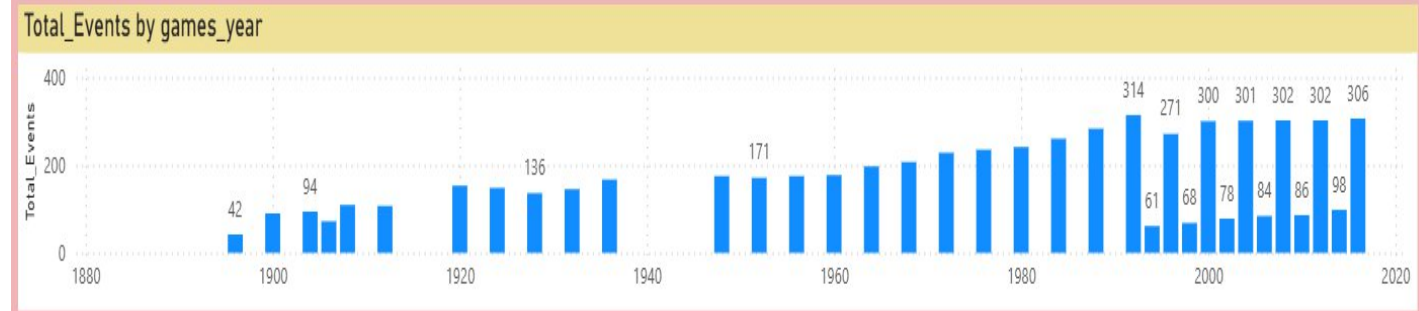
9. Events Over Time: Column chart visualizing Total Event measure by Games Year.

city_name	Total_Games
Athina	3
London	3
Innsbruck	2
Lake Placid	2
Los Angeles	2
Paris	2
Sankt Moritz	2
Stockholm	2
Albertville	1
Amsterdam	1
Antwerpen	1
Atlanta	1
Total	51

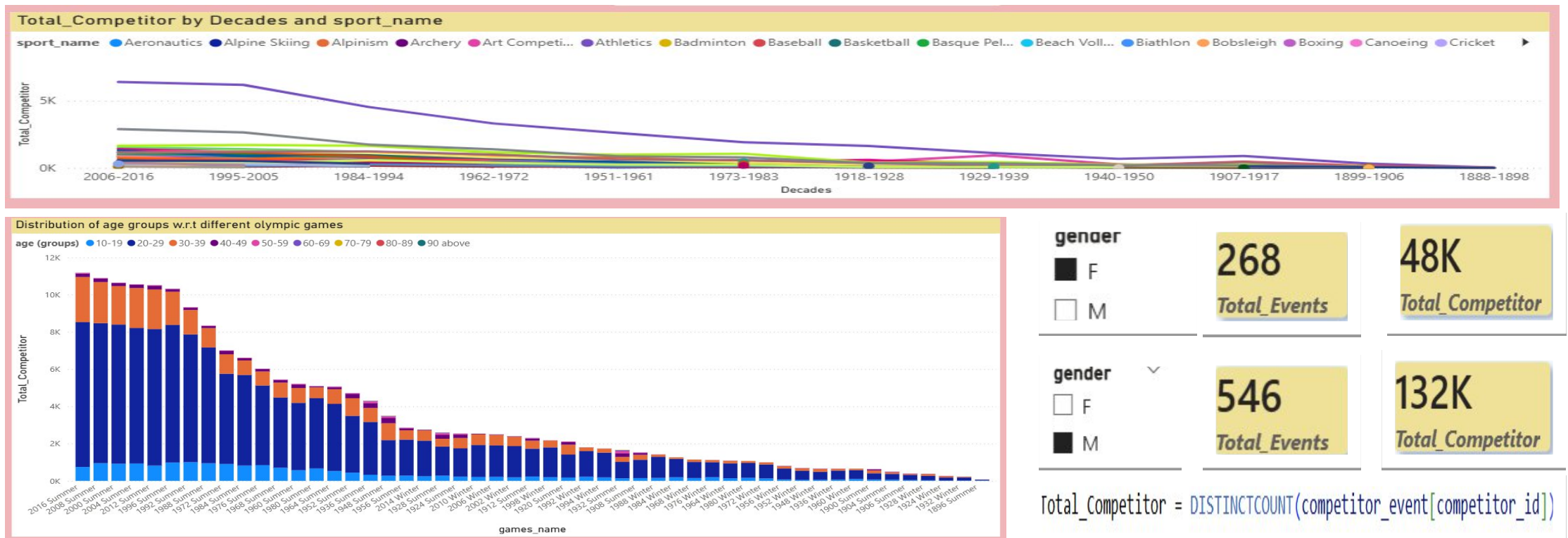
sport_name	Total_Events
Athletics	83
Shooting	83
Swimming	54
Cycling	44
Sailing	37
Wrestling	30
Art Competitions	29
Canoeing	27
Gymnastics	27
Archery	25
Cross Country Skiing	23
Rowing	23
Total	757

```
Total_Games = DISTINCTCOUNT(games[id])
```

```
Total_Events = DISTINCTCOUNT(event[id])
```



12. Age Distribution: Clustered column chart (Total Participants by Game Name, legend = Age Group 10-19, 20-29, 30-39).



Power BI: Country Performance & Medals

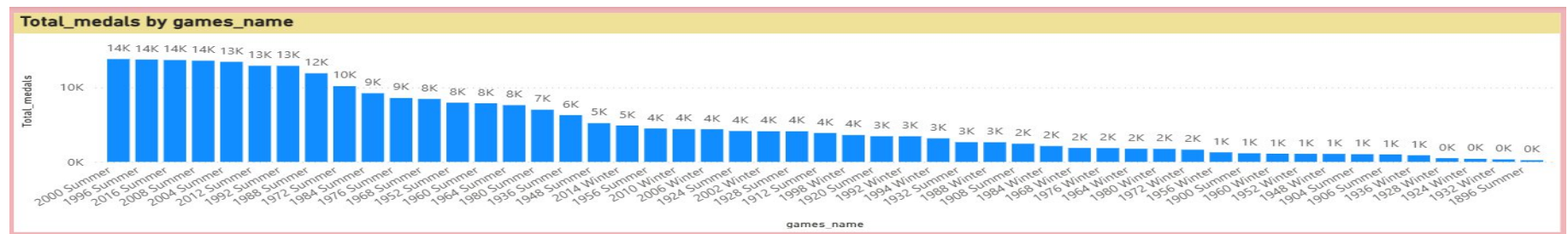
11. Highest Participants by Country: USA leads with 12,392 participants (Table visual).

13. Medals Per Olympics: Column chart with data labels enabled (Total Medals by Games Name).

14. Highest Gold Medals: USA received the highest gold medals (2,533) (Table visual with Total Gold Awarded measure).

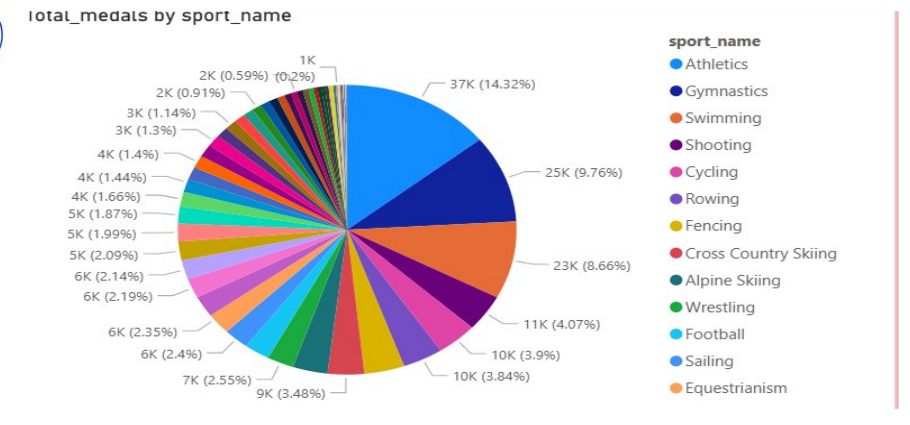
15. Medals by Sport: Pie chart showing Total Medal distribution across Sport Names.

region_name	Total_Competitor
USA	12392
UK	7999
France	7323
Germany	7119
Italy	6958
Canada	6332
Australia	5452
Sweden	5416
Japan	5355
Soviet Union	4263
Poland	4241
Hungary	4045



Total_Gold_Awarded = CALCULATE(COUNT(competitor_event[medal_id]),competitor_event[medal_id]=1)

region_name	Total_Gold_Awarded
USA	2533
Soviet Union	1135
Germany	843
UK	658



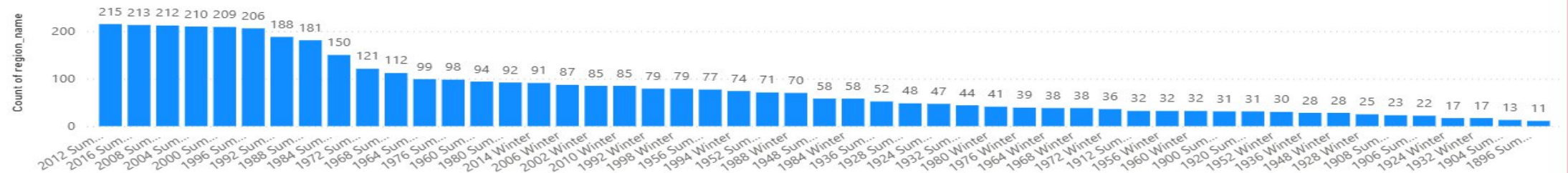
Power BI: Regional Analysis

16. NOCs Per Game: Column chart showing count of region names by Games Name.

17. Highest Participant Regions: USA leads (Table visual sorted descending).

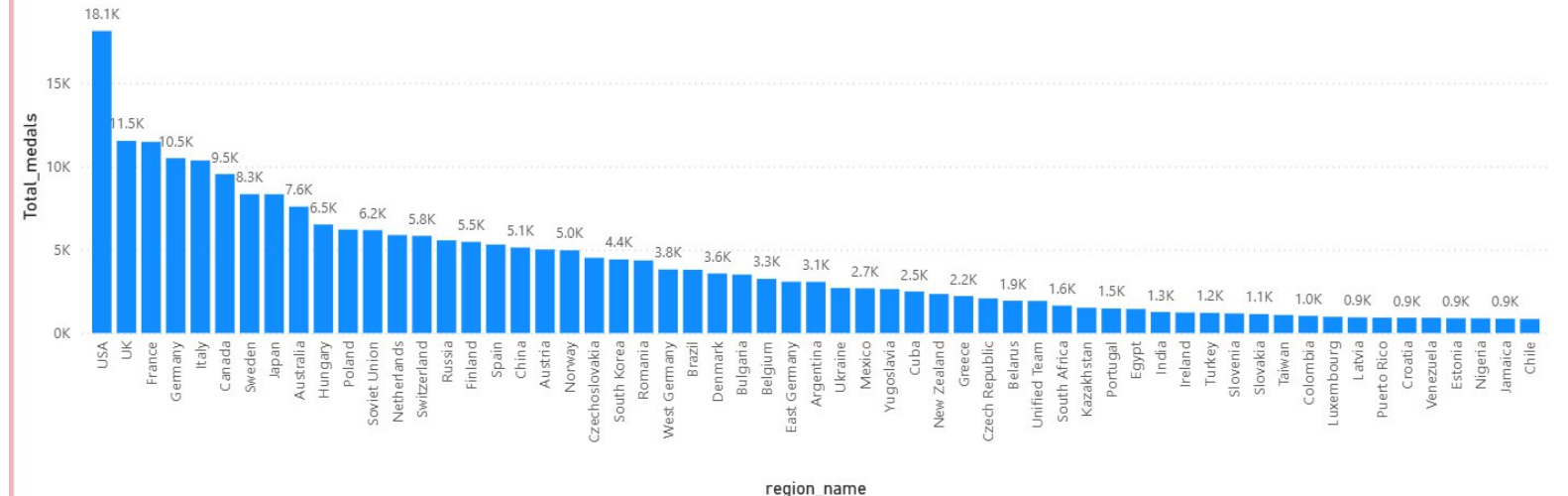
18. Medal Distribution by Region: Column chart visualizing Total Medals by Region Name.

Count of region_name by games_name



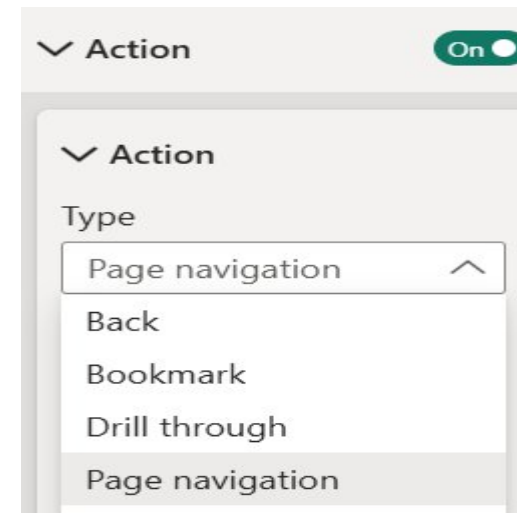
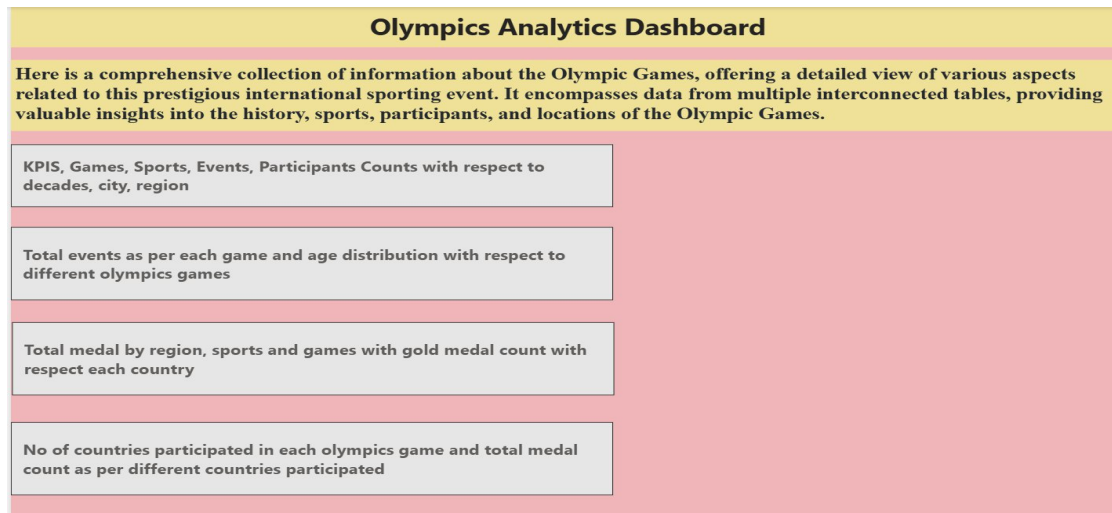
region_name	Total_Competitor
USA	12392
UK	7999
France	7323
Germany	7119
Italy	6958
Canada	6332
Australia	5452
Sweden	5416
Japan	5355
Soviet Union	4263
Poland	4241
Hungary	4045

Total_medals by region_name



Final Interactive Dashboard Design (Power BI)

- **Data Modelling:** Managed relationships, made all connections active, enabled bi-directional filtering (Dimension <-> Data Table).
- **UI/UX:** Title box inserted ('Olympics Analytics Dashboard'), formatted for visual appeal.
- **Navigation:** 4 action buttons inserted for page navigation. Action set to 'Page navigator' linking to relevant visual pages. Text enabled on buttons for context.
- **Alignment:** All KPIs, charts, graphs, and trends aligned properly.
- **Interactivity:** CTRL + Click on buttons navigates to respective dashboard pages seamlessly.



CTRL+1st action button

Total_Games**Total_Sports****Total_Events****Total_Competitor**

Total_Region

Total_City

Total_medals

region_name	Total_Competitor
USA	12392
UK	7999
France	7323
Germany	7119
Italy	6958
Canada	6332
Australia	5452
Sweden	5416
Japan	5355
Soviet Union	4263
Poland	4241
Hungary	4045
Total	180252

gender 

☐ F

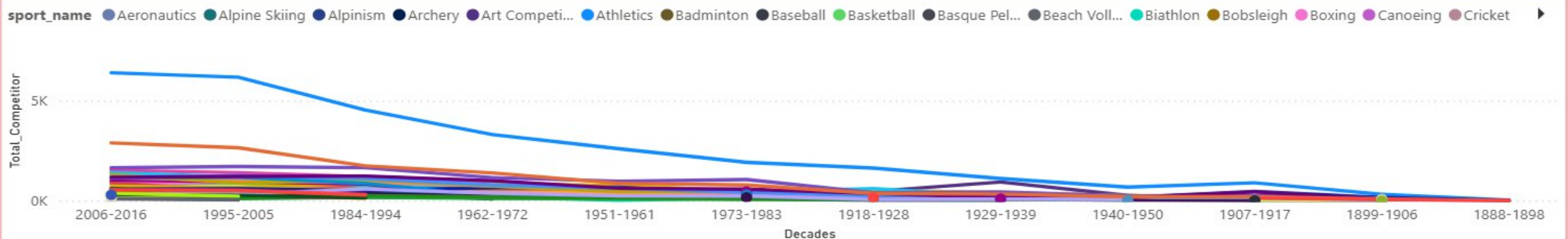
☐ M

season 

☐ Summer

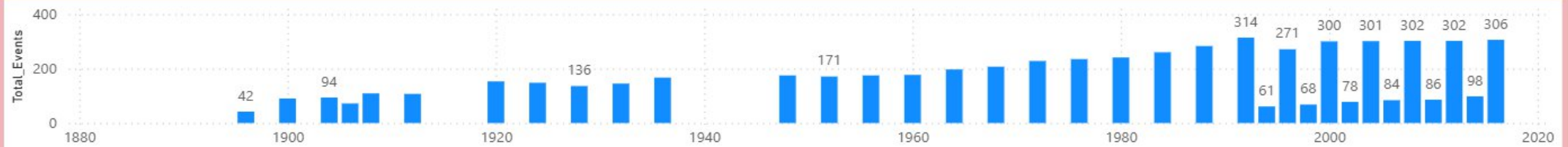
☐ Winter

Total_Competitor by Decades and sport_name



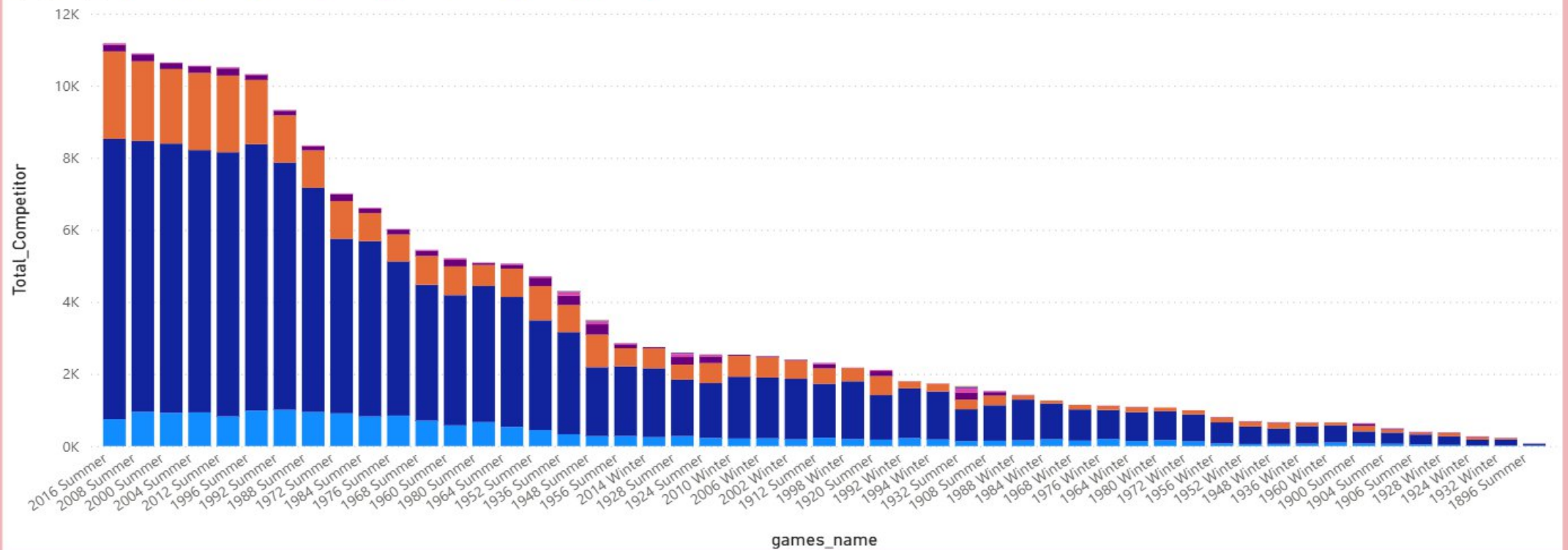
CRTL+2nd action button

Total_Events by games_year



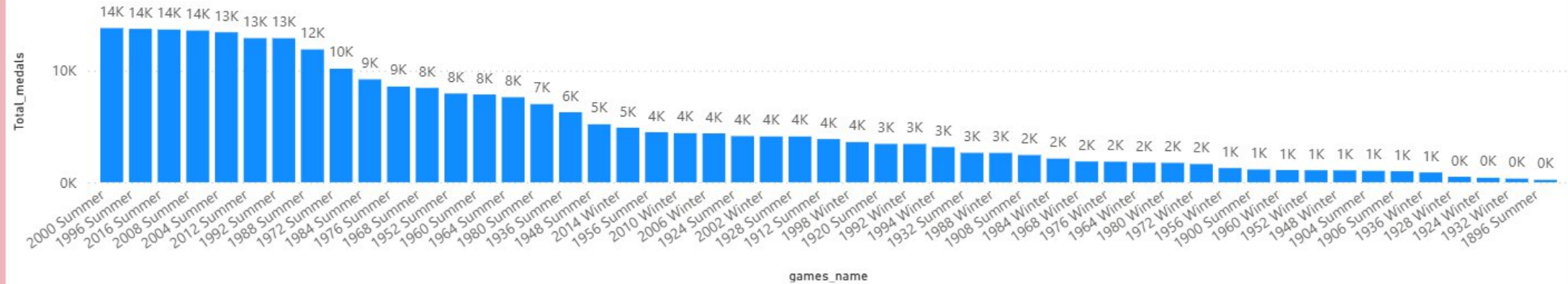
Distribution of age groups w.r.t different olympic games

age (groups) ● 10-19 ● 20-29 ● 30-39 ● 40-49 ● 50-59 ● 60-69 ● 70-79 ● 80-89 ● 90 above



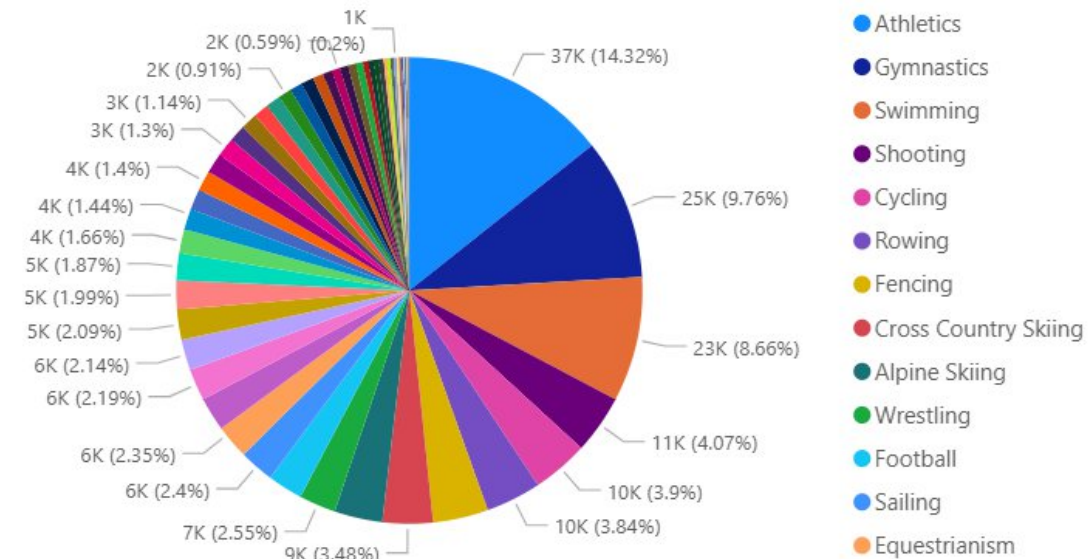
CRTL+3rd action button

Total_medals by games_name



region_name	Total_Gold_Awarded
USA	2533
Soviet Union	1135
Germany	843
UK	658
Italy	568
France	492
Sweden	477
Russia	466
Canada	461
East Germany	451
Hungary	434
Norway	378
China	350
Australia	348
Total	13106

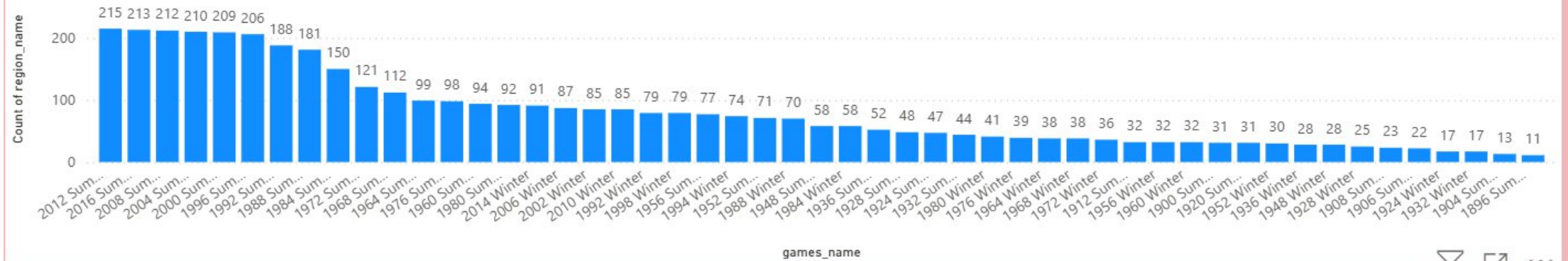
Total_medals by sport_name



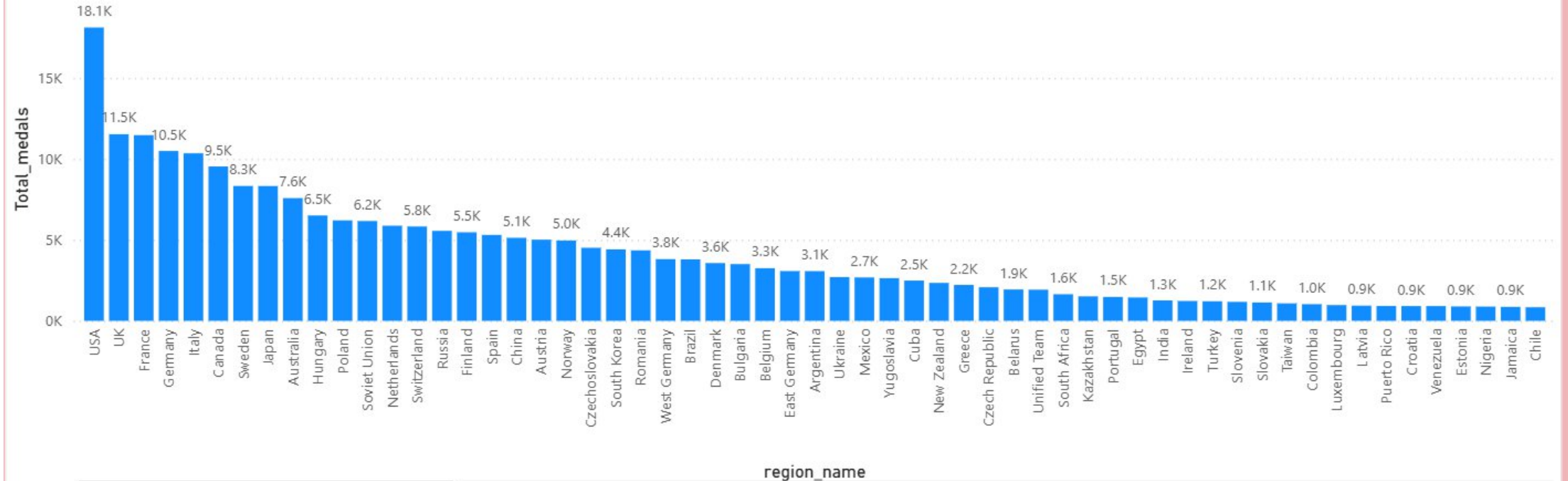
medal_name
☐ Bronze
☐ Gold
☐ NA
☐ Silver

CTRL+4th action button

Count of region_name by games_name



Total_medals by region_name



Conclusion

- The Olympic Games have evolved significantly in scale, gender inclusion, and global representation.
- Geographical and cultural factors strongly influence sporting dominance (e.g., India in Hockey, Scandinavia in Winter Sports).
- Athlete physical demographics have trended upward, reflecting modern training and nutrition.
- Interactive BI Dashboards allow for seamless, dynamic exploration of these historical trends.